

Can LLMs be used to Quantify the Emotional Salience of Text Statements using an Elo Rating System?

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Abstract

When studying mood or affective state it is useful to collect free-text input from participants relevant to their mood alongside traditional assessment batteries. Quantifying free-text data in relation to quantitative measures remains a challenge. We propose a novel application of an Elo rating system for analysing verbatim human text descriptions of emotionally salient experiences. By leveraging crowdsourced pairwise comparisons, our approach preserves the richness of free-text data while generating a quantitative representation. We apply the same approach to various LLMs to compare their performance against humans. Regression analyses indicate that rankings generated by LLMs significantly predict human rankings ($p < 0.001$), demonstrating strong explanatory power across models ($R^2 = 0.760\text{--}0.803$). Applying LLMs to this task offers the added advantages of greater efficiency and lower cost. We discuss the potential applications of this method in human research and explore critical considerations regarding the use of LLMs in emotionally relevant tasks.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

Large language model, Emotion Research, Pairwise Comparisons, Elo Rating, Crowdsourced, Life Events

*Both authors contributed equally to this research.

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1 Introduction

A “subjective construct” can be defined as a narrative derived from personal feelings, opinions, and experiences with no objective ground truth, for example, toxicity of online content [3] or emotional experiences [21]. Understanding and quantifying emotional experiences are crucial for research in areas such as mental health and human-computer interaction [1, 10, 25]. Subjective insights enable researchers to capture nuanced aspects of human behaviour that are often overlooked in traditional assessment tools. Mixed-methods researchers collect information on these constructs in both quantitative and qualitative formats, producing free-text responses linked to numerical measures. Combining subjective and quantitative data in emotion research enables the development of diagnostic criteria, the creation of mathematical models, and the tracing of causal relationships using statistical tools. However, analysing free-text data is challenging. Manual methods, such as thematic analysis [9], are often time-consuming and resource-intensive, while automated approaches may fail to account for contextually significant features in the data. For researchers without the expertise or time to conduct detailed qualitative analyses on a small subset of their dataset, alternative approaches to efficiently quantify free-text data are beneficial.

Elo rating systems (ERS) [14] are traditionally used in zero-sum games, such as chess, to give an overall ranking of players based on the outcomes of two-player games. This is useful because pairwise comparison is an effective method for evaluating items; the pairwise comparison method [13] involves making a judgement about which of two items is better or has more of a specific property. By constraining the judgement to just two items, this method reduces

cognitive load for participants and decreases response times compared to alternate questionnaire methods [11, 27, 38]. ERS have no limit on the number of items that can be ranked, and are designed to accommodate variability [23], making them a valuable tool for subjective data analysis. However, when applied to free-text data of indeterminate length, human factors such as reading speed and fatigue can still pose issues to the scalability of this approach [37].

Recent advances in LLMs offer promising solutions for analysing free-text data at scale [23, 34]. Their ability to process and quantify text efficiently opens new avenues for this type of research, although it remains critical to acknowledge their performance limitations, especially in tasks pertaining to the subjectivity of lived experiences in real-world contexts [44]. However, if LLMs can interpret the emotional significance of human-generated text in a manner comparable to humans, they could complement analysis by providing consistent and scalable evaluation tools, while acknowledging the complexity and variability inherent in human experiences.

To investigate this, we developed a crowdsourced ERS to quantify the relative score / rank position of previously collected descriptions of “significant events”. To our knowledge, previous researchers using these methods for crowdsourced subjective free-text data analysis perform the pairwise comparison ERS method after data collection has concluded by computing the Elo rank changes from an amalgamation of all pairwise comparisons [27, 38]. In contrast, this paper presents a live-time updating approach which allows rankings to evolve dynamically after each comparison is made, simultaneously across multiple user inputs. We then tested whether LLMs could perform the same task, and if so, to what level of accuracy. Overall, this paper provides:

- a true-to-life (verbatim) dataset of daily lived experiences that impact human mood states,
- a methodological contribution in the form of a novel live-time application of an ERS for the purpose of crowdsourced quantification of subjective free-text data, and
- a comparison of human and LLM performance on this task, accompanied by a discussion of future developments for this work.

2 Related Work

We place our work at the cross-section of previous work engaged in answering “*how can we quantify subjective free-text data?*” and “*can LLMs show empathic understanding of human lived experiences?*”. We critique the use of synthetic free-text datasets and discuss the utility of LLMs for empathic judgement tasks.

2.1 How can we Quantify Subjective Free-Text Data?

Aggregating crowdsourced opinions of inherently subjective constructs is a useful method to convert diverse beliefs into an average benchmark or “inter-subjective ground truth” [27, 43]. Grouping statements into predefined categories is a common approach many researchers use, for example to classify social media posts into depression severity levels [28], or to derive stressor categories from SMS-like sentences [26]. Similarly, tools like Automatic Histograms [36] and EmPath [15] analyse free-text content by quantifying the

presence and strength of lexical categories, producing a numerical representation through a purely summative approach. While these methods effectively depict features such as emotional tone, they fall short of capturing the full depth and connotations of free-text statements. As such, directly assessing free-text data using pairwise comparisons with the “majority-vote” method has been previously used to decide “ground-truth” answers, however this method is vulnerable to inflating bias and random error [27]. Alternatively, pairwise comparisons with Elo rating enables ranking of subjective data while preserving its richness with a method that is unsusceptible to ceiling effects and removes systematic selection bias due to participant fatigue [38]. Moreover, ERS methods are well-suited for diverse study designs, including longitudinal studies, as they accommodate variable sample sizes and dynamic data changes while maintaining ranking consistency [30].

2.2 Can LLMs show Empathic Understanding of Human Lived Experiences?

In order to make judgements on emotionally significant lived experiences a certain level of emotional intelligence (EI) is required. In humans, we conceptualise EI as the ability to recognise, apply, understand, and regulate emotions [39]. LLMs can already emulate many aspects of EI and there is a growing literature on the necessity for empathic computing in human-computer interaction [41]. Empathic understanding - the ability to recognize and connect with another person’s thoughts, feelings, and experiences - is fundamental for interpreting human emotions and perspectives. Recent advances suggest that conversational LLMs may exhibit social skills and empathic inference comparable to humans in specific contexts [22, 42, 45]. Notably, a systematic review has highlighted that LLMs exhibit multiple elements of human-like empathic understanding, suggesting their potential to mimic this critical aspect of EI [40]. However, LLM performance may still have limitations compared to human capabilities, particularly in complex or nuanced emotional tasks [12, 20].

A potential issue with previous research in this area is the use of artificially created datasets of emotionally significant items. For example, [42] constructed persona scenarios designed to include a mixture of emotional terms within a pre-specified format. They then introduced further researcher bias by selecting four of the “most plausible emotions” for each scenario to be rated with by both humans and LLMs [42]. Similarly, [20] compiled a dataset of emotion-eliciting scenarios to evaluate empathy in both humans and LLMs, by aggregating fictional situations used in prior human research studies. While these approaches are useful, they rely on scenarios that may not fully reflect real-life emotional contexts. While datasets like EmpatheticDialogues [35] are emerging as more authentic verbatim sources, many researchers still rely on manufactured scenarios. This introduces two key challenges. Firstly, using artificial rather than real statements when assessing AI models can result in evaluations that fail to reflect the richness and diversity of real-world emotional experiences, leading to less reliable assessments of model performance. Secondly, when applying AI to tasks such as ranking qualitative data, artificial statements may not adequately capture the variability and interpretative breadth inherent in real input. Recognising and addressing these distinctions

in the materials used to evaluate model performance could have significant implications, particularly if these models are applied in sensitive real-world settings such as therapy, where their ability to handle genuine, complex human experiences is paramount.

3 Method

3.1 Significant Events Dataset

Between July 2023 and March 2024 we conducted three online research studies [5–7] which involved collecting immediate mood state assessments from a total of 1275 human participants. Within the mood questionnaires we asked participants *“Is an event or experience (either positive or negative) significantly affecting your mood right now?”* with a yes/no response option. If participants answered “yes”, they would be asked to *“Please indicate on the sliding scale how positive or negative the experience was for you”* with a sliding scale response option ranging from 0 (Negative) to 100 (Positive), and then asked *“If you feel comfortable, please briefly describe the event/experience that is significantly affecting your mood right now”*. See Appendix A.2 for full Significant Event Questionnaire used in the original studies. 437 participants reported an event significantly impacting their mood (322 negative, sliding scale score < 50; 113 positive, score > 50; and 2 neutral, score = 50). It is important to note that whether an event is referred to as “positive” or “negative” is solely based on the participant’s subjective rating at the time of describing the event. Participants had up to 500 characters to detail the event (hereafter referred to as “statements”) and could refrain from answering this question by inputting arbitrary characters into the text box. As such, the statements collected were of various lengths and ranged in level of detail (see Appendix A.3, Figure 3 for further details). Minimal data cleaning procedures were applied to the statements to maintain their true-to-life nature, the full cleaning procedure is detailed in Appendix A.4. After cleaning, 363 statements remained (265 negative; 98 positive).

3.2 Elo Rating System

Here, we describe a standard ERS as applied to ranking statements, analogous to its common use in ranking chess players, where statements represent players and comparisons represent games. We adopt chess terminology, such as a statement “winning” a comparison to mean being judged as more positive. The initial Elo value (E_0) for a statement (S_1) was calculated based on the participant’s own reported statement value $V(S_1)$ (i.e. the sliding scale response from 0–100), as follows:

$$E_0(S_1) = \lfloor (1000 - 50 \times 2.5) + V(S_1) \times 2.5 \rfloor. \quad (1)$$

This adjusts the centre-point to 1000 and places $E_0(S_1)$ onto a starting scale of 875–1125. The ERS then uses the following equation to estimate the probability of a statement winning after a pairwise comparison:

$$P(S_1) = \left[1 + 10^{\frac{E(S_2) - E(S_1)}{400}} \right]^{-1} \quad (2)$$

where $P(S_1)$ is the estimated probability of S_1 winning and $E(S_1)$ is the Elo score of S_1 . Similarly, $P(S_2) = 1 - P(S_1)$. Based on the result of the comparison, Elo scores are then updated using a typical Delta

rule-based update:

$$E'(S_1) = E(S_1) + \lfloor K \cdot (\delta(S_1) - P(S_1)) \rfloor \quad (3)$$

where $E'(S_1)$ is the new Elo score for S_1 , $\delta(S_1) = 1$ if S_1 wins, and zero if S_1 loses¹. Here $K = 32$, this is standard choice in traditional ERS for novice players [16], which will allow for initial larger rank movements within our scale.

3.3 Pairwise Comparison Method

Hitherto, reference to “participants” reflected the participants from the previous mood research studies who contributed the significant event statements. Hereafter, the online participants in this Elo ranking study will be referred to as “participants in the Elo ranking study” or “PERSs”.

A pairwise comparison method was used in combination with the ERS. In this method, two statements were shown with instructions asking *“In your opinion, which of these scenarios is BETTER?”*. Question phrasing as “BETTER” or “WORSE” was counterbalanced to prevent influencing PERSs or LLM interpretation of the valence of a statement. 50 pairwise comparisons were completed per PERS / LLM session. To generate pairs of statements for comparison, statement one (S_1) is selected at random from the full list of statements. Statement selection is designed to ensure each statement is seen an equal number of times. S_1 is selected from the statements that have fewer comparisons, S_2 is then selected from the statements ± 10 rank positions surrounding S_1 , with probabilities based on normal distribution so closer statements are favoured. The same pair of statements was never shown more than once to any PERS but they could be shown the same individual statement more than once. After a comparison was made, the loser statement, “moved down” in the rankings, and the winner statement “moved up” according to the ERS.

3.4 Live-Time Rank Updating

For the human version of this study, to optimise data collection efficiency, multiple PERSs could perform the study simultaneously, with up to ten PERSs at any given time. To enable live updating of Elo scores while ensuring useful and valid comparisons, the process had to be compatible with the automated integration of PERSs responses. Specifically, once a statement was presented to a PERS, it was temporarily unavailable to others. Only after a pairwise comparison was completed and the Elo scores for the respective statements were updated would they become available for presentation to another PERS.

3.5 Study Design

3.5.1 Human Study Design. The website for the human version of this study was hosted on a private server managed by C.H. The complete human study procedure is outlined in Figure 4 in Appendix A.5, which also includes an example of the pairwise comparison question format. PERSs signed up for the study, reviewed the information sheet, and provided electronic confirmation of informed consent. Only those who passed the comprehension check were allowed to participate in the remainder of the task.

¹In some application $\delta(S_1) = 0.5$ for a draw but that is not a possible outcome here

3.5.2 Human Participants in the Elo Ranking Study. We recruited 142 PERSs through Prolific.com[33] between 2024-10-28 and 2024-10-30. The target sample size was 140 PERSs, however two PERSs completed the study exceptionally quickly (under 4 minutes), so two additional PERSs were recruited. Eligibility criteria required PERSs to be aged 18-30 years, fluent in English, currently living in the UK, not currently taking medication for a psychiatric disorder, and not previously involved in a study run by K.B on Prolific (criteria matching those used in the prior studies that generated the statements). PERSs were reimbursed 1.91GBP for completion, with a median study duration of 10 minutes and 55 seconds. Ethical approval was obtained from the Research Ethics Committee of the University of Bristol (School of Psychology, Ethics approval code 21823).

3.5.3 LLM Study Design. Three LLMs were chosen to complete the task in addition to humans; GPT-3.5-Turbo [32], Claude-3.5-Haiku [2] and Gemini-1.5-Flash [17]. The models were run between 2024-11-25 and 2024-12-09 for a minimum of 140 sessions, where each session can be thought of as a simulated PERS. This allowed us to collect a minimum of 38 comparisons for each statement; this matches the available data collected for the human PERSs. A new LLM session was started every 50 pairwise comparisons to clear memory to emulate novel human PERSs. As such, if the models work perfectly, 7000 pairwise comparisons are required to achieve 38 comparisons for each of the 363 statements. The main alteration to the study code to ensure LLM compatibility involved removing the live-time rank updating functionality used in the human version. Live-time updating was not necessary for the LLMs as one pairwise comparison is completed at a fraction of the speed a human takes to perform the same task and there is no waiting time for PERSs to sign up to the study, therefore each session of 50 comparisons could be performed sequentially. A modified version of the human study information was given to the LLMs to constrain the output to select only the first or second statement in the pairwise comparison. For each pairwise comparison the LLM question prompt format used was: “Which of the following life experiences is comparatively better? (Statement 1). (Statement 2). Please respond with only ‘1’ or ‘2’ without any additional text.”. We also created an agent which selected at random from the binary answer options (S_1 or S_2). This was used as a baseline measure to reflect the performance expected with no understanding or reasoning about the task.

4 Results

To illustrate the Elo rating method, Figure 1 shows the Elo score trajectories of 25 statements, from human (a) and GPT (b) raters. For clarity, Figure 1 plots the trajectories of statements with the same initial Elo score $E_0 = 875$. The pairwise comparison method redistributes these tied scores along a continuous scale. In this subset, the lowest-ranked statement–222: “death; cheating”–ends up ~400 points below the highest-ranked statement–155: “Moving house and dealing with the stress of packing”. This highlights how the ERS method differentiates statements based on relative score.

For convenience $E_{t,m}$ is used for the Elo score after t comparisons using model m ; with $m = h$ for the human ranking. A simple linear regression was conducted to examine whether E_0 predicted $E_{38,h}$. The results showed a significant positive relationship, $\beta = 1.24$,

$F(1,361) = 20.38$, $p < .001$, with a 95% confidence interval of [1.12, 1.36]. The model suggests that the E_0 explained 53.5% of the variance in $E_{38,h}$ ($R^2 = .535$).

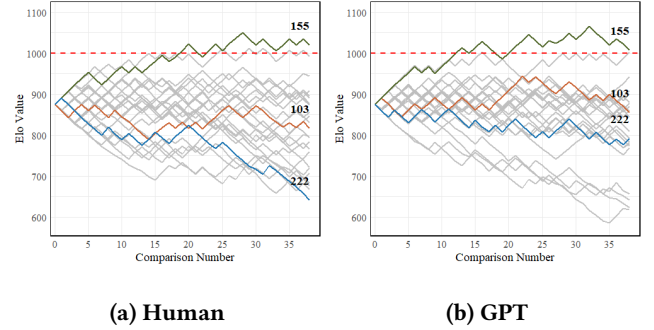


Figure 1: Elo Score change over time in human (a) and GPT model (b) versions of the study as a result of pairwise comparisons. The 25 statements plotted in this example have a tied start Elo score of 875. For the human group, the minimum, maximum, and median of the 38th Elo scores are highlighted with their corresponding statement ID labelled. The x -axis plots the number of pairwise comparisons per statement, and the red dotted line at 1000 on the y -axis represents the “neutral” line. Please note, while this subset of statements is visualised for clarity, their rankings still evolved as part of the larger dataset encompassing all 363 statements. 155 = “Moving house and dealing with the stress of packing”, 103 = “I lost a friend.”, and 222 = “death; cheating”. Please note, this group exemplar is just a subset of the full data to aid visualisation of these results, all 363 statements are detailed in Figure 5 in the Appendix.

Multiple linear regression analyses were used to assess whether the LLMs predict the human ranking. For all three LLMs we find $E_{38,m}$ significantly predicts $E_{38,h}$ ($ps < .001$), controlling for the initial value E_0 : GPT model $R^2 = .788$; Gemini model $R^2 = .760$; and, Claude model $R^2 = .803$. The random agent’s E_{38} was not a significant predictor of $E_{38,h}$ ($p = .241$), any variance explained in the random agent model was only due to E_0 , with $R^2 = .537$. See Appendix A.8, Table 1 for the full model outputs.

Figure 2a plots the variance explained (R^2) by the regression model ($E_{38,h} \sim E_0 + \beta_1 \cdot E_{t,m}$) for each t for the five different versions: Human, Random, Claude, GPT, and Gemini. The x -axis shows t and the y -axis the corresponding R^2 . Figure 2b illustrates the correlation between the human and GPT Elo scores after 38 pairwise comparisons per statement: the results indicated a strong positive pearson’s correlation, $r(361) = .88$, $p < .001$, with a 95% confidence interval of [.855, .902]. For further examples, the three best and worst-ranked statements after 38 comparisons for human and GPT models can be found in Appendix A.8, Table 2.

To confirm that 38 comparisons per statement was sufficient to minimise order-dependency effects inherent in Elo systems [11], mElo scores were calculated using the “EloChoice” package [29] with the human data only. The original pairwise comparison sequence resulted in the $E_{38,h}$ scores used in the results reported above.

mElo scores represent the mean of each statement’s Elo score after 99 randomisations of the pairwise comparison sequence and the original sequence. Pearson’s correlation revealed a strong positive relationship between $E_{38,h}$ and mElo, $r(361) = .94$, $p < .001$, with a 95% confidence interval of [.921, .947] (akin to Clark and colleagues Elo-mElo correlation of 0.95 [11]).

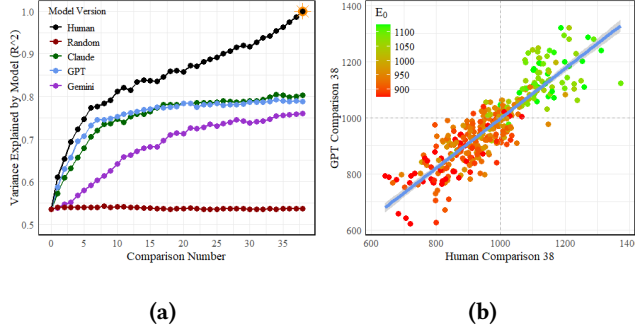


Figure 2: (a) The variance explained for each model. Orange star highlights the human model comparison 38 as this is the dependent variable in the models. (b) Pearson’s correlation between human comparison 38 and GPT comparison 38. Colour-coded by the statement starting Elo score, where higher values reflect statements that started with more positive ratings. Dashed grey lines at 1000 represent the neutral centre of the scale.

5 Discussion

Our experiment demonstrates that this is a useful methodological approach for creating a crowdsourced quantitative scale for subjective free-text statements. We show that LLMs can perform the task in a similar manner to crowdsourced raters, who serve as the ground truth for these evaluations. Although differences between LLM and human performance highlight distinct approaches rather than a clear advantage for either, using LLMs for these tasks offers significant time and cost benefits. For example, the human version of this study cost 379.71GBP and took ~9 hours to collect data from 142 participants, whereas the GPT model performed the task in ~45 minutes for ~6.47GBP. The three LLMs used in this study were selected for affordability rather than optimal performance, positioning our results at a lower performance boundary and highlighting the method’s potential. The ease of using LLMs for this task makes them a valuable tool for analysis with narrative data and highlights potential uses for empathic computing. The live-time updating method of statement selection in the human version of this study could be easily adapted to incorporate new statements into the dataset without restarting the study, just as a new chess player can be assigned a rating after only a few matches without competing against all existing players. This could allow statements to be ranked in real time during data collection, for example, in a mood-monitoring app.

Our method is particularly useful when free-text data is part of a larger dataset that includes related quantitative measures. For

example, the dataset these statements originate from contains cognitive assessments, mood questionnaires and other significant event questions in quantitative formats. Converting free-text responses into numerical data allows integration with statistical analyses and also generates further research questions when there are apparent discrepancies between participants’ subjective scale ratings and crowdsourced PERSS’ ratings. This demonstrates the value of our method as a tool for contextually relevant free-text quantification, especially if, like us, you wish to use a quantitative measure collected alongside the free-text as the starting Elo score. While this method could also be applied as a stand-alone free-text ranking approach—for example, to generate a ranked list of life events from best to worst—its utility will largely depend on the intended use of the crowdsourced data. Therefore, careful consideration of the Elo starting point is crucial in study design to ensure alignment with research objectives.

5.1 Limitations and Future Work

The statements used in this study were collected from prior research studies with 18–30 year olds. To maintain demographic consistency, the human raters in this study were also aged 18–30, ensuring event comparisons were contextually relevant. For example, an event like “death of a sibling” may be more unexpected for someone aged 18–30 than for someone aged 80–90. This design prioritises demographic specificity over generalisability but could be easily replicated with other groups or subpopulations.

Accordingly, this raises the question, “*What demographic do LLMs embody when making these comparisons?*”. Mainstream LLMs, primarily trained on the CommonCrawl dataset, reflect biases towards English-speaking, younger males from WEIRD societies [4, 18, 19, 24, 31]. Differences in knowledge cut-off dates may also explain discrepancies in model responses. For instance, the statement “*new regulations regarding UK visa - worrying about my future*” referring to changes announced in the UK in December 2023, would be unknown to GPT-3.5-turbo (September 2021 cut-off), but likely accessible to newer models with later cut-off dates like Gemini 1.5 Flash (November 2023) and Claude-3.5-Haiku (July 2024). Although online research participants are at least more representative than in-person convenience samples [8], we must acknowledge that the alignment between LLM and human ratings observed here could reflect an overlap between our human sample and LLM training data. This raises important questions around how this method may be applicable to under-represented populations which warrants further research.

Next steps include: (1) Integrating the new crowdsourced quantified data back into the original datasets to compare ‘self’ versus crowdsourced perceptions in studying mood’s impact on cognitive processes; (2) Experimenting with different LLM prompts or personalised GPTs to see if framing the model as a specific demographic affects outcomes; (3) Assessing whether newer, more advanced models with improved ‘reasoning’ achieve closer-to-human performance.

6 Conclusion

This study introduces a novel application of live-time ERS for crowd-sourced quantification of subjective free-text data, providing a scalable and efficient method to rank lived experiences. By demonstrating that LLMs can perform this task to a similar level as humans this paper highlights new potential for leveraging AI to streamline this type of analytic approach for mixed-methods research. However, limitations such as demographic biases in the participant sample and LLM training data underscore the need for careful consideration when applying this method to diverse populations. The integration of real-time ranking presents important implications for designing interactive systems that can dynamically adapt to user-generated content, offering practical applications in mental health, user experience research, and adaptive human-centred technologies.

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A Appendix

A.1 Software and Data Statement

The human and LLM code (including model parameters) used in this research is available in a Github Repository: [github/Elo-Project](https://github.com/Elo-Project).

A.2 Significant Event Questionnaire

The following question was asked to assess the impact of a significant event on a participant's mood state:

- (1) *"Is an event or experience (either positive or negative) significantly affecting your mood right now?"* - Binary response options given: 'Yes' / 'No'.

If participants responded **Yes**, they were then asked four further questions:

- "Please indicate on the sliding scale how positive or negative the experience was for you:"* - Sliding scale response option ranging from 0 (Negative) to 100 (Positive).
- "How long ago did the event happen?"* - Three button response choices: 'Today', 'Yesterday', '2+ Days ago'.
- "Did you know the event was going to occur before it happened?"* - 4 button response choices: 'No – it was a complete surprise', 'Yes – on the day it happened', 'Yes – up to a week before it happened', 'Yes – I knew about it for more than a week before it happened'.
- "If you feel comfortable, please briefly describe the event/experience that is significantly affecting your mood right now."* - Optional free-text box response.

A.3 Significant Event Statement Length

The frequency distribution of the statement character length is shown in Figure 3.

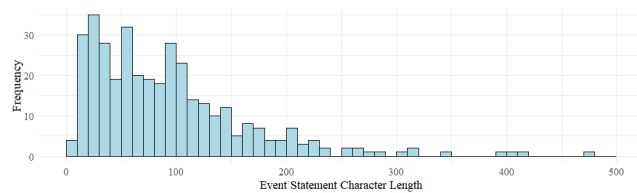


Figure 3: Frequency distribution plot of significant event statement character length with binwidth of 10. Range = 6 to 476, Mean = 93, Mode = 24.

A.4 Significant Event Statement Cleaning Procedure

The following cleaning procedure was applied to the 437 records of significant events:

- Removal of blank or "N/A" type statements (N=52)
- Removal of "I do not feel comfortable sharing" type statements (N=11)
- Removal of duplicated statements from the same participant (restarting the study) (N=8)
- Removal of incomplete statements (N=1)
- Modification of statements containing personally identifiable information (N=1)
- Modification of statements referencing specific days of the year that would change the meaning of the statement at the point of use in the task (N=4)
- Correction of spelling errors and/or removal of coding syntax such as "\n" (N=56)

A.5 Human Study Procedure

The study flow diagram as outlined for the human rater task is detailed in Figure 4.

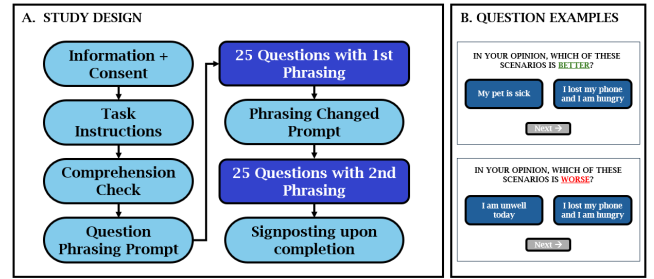


Figure 4: (A) Study procedure flow diagram detailing the order that human participants completed the task. "Phrasing" here refers to whether the Question asked participants to select the "better" or "worse" statement from the pair. (B) Example question screen layouts (not using actual statements) to show the two question phrasing prompts.

A.6 Further Details of Methods

The human version took ~9 hours to collect data from 142 PERs. In contrast, GPT-3.5 performed the task in ~45 minutes, and Claude in 2 hours and 20 minutes due to a limit of 50 requests per minute (RPM), where one request = one pairwise comparison. Gemini took ~5 days to complete as the free model has usage limits of 15 RPM and 1500 RPD (requests per day).

A.7 LLM Prompt To Explain The Purpose Of The Study

"You are invited to take part in a research study to compare statements about life experiences.

What is this study about?

The purpose of this study is to better understand judgments of life experiences. The life experience statements used in this study vary

widely. Some statements may refer to everyday occurrences (for example, losing an item); other statements may refer to more impactful life events (for example, loss of a loved one). By asking you to compare these differing types of statements, we aim to understand how you rank various life experiences.

What will you have to do?

You will be asked to compare multiple pairs of statements that refer to life experiences, one pair at a time. You will have to decide which of the statements is comparatively better or worse than the other, depending on the phrasing of the question when these statements are presented to

you. The events are provided without context, so you can be as general as you wish.

Please respond with only '1' or '2' without any additional text."

A.8 Detailed Results Output

Figure 5 provides a full overview of the 363 statement paths. Table 1 details the full model outputs from the regression models predicting human comparison 38.

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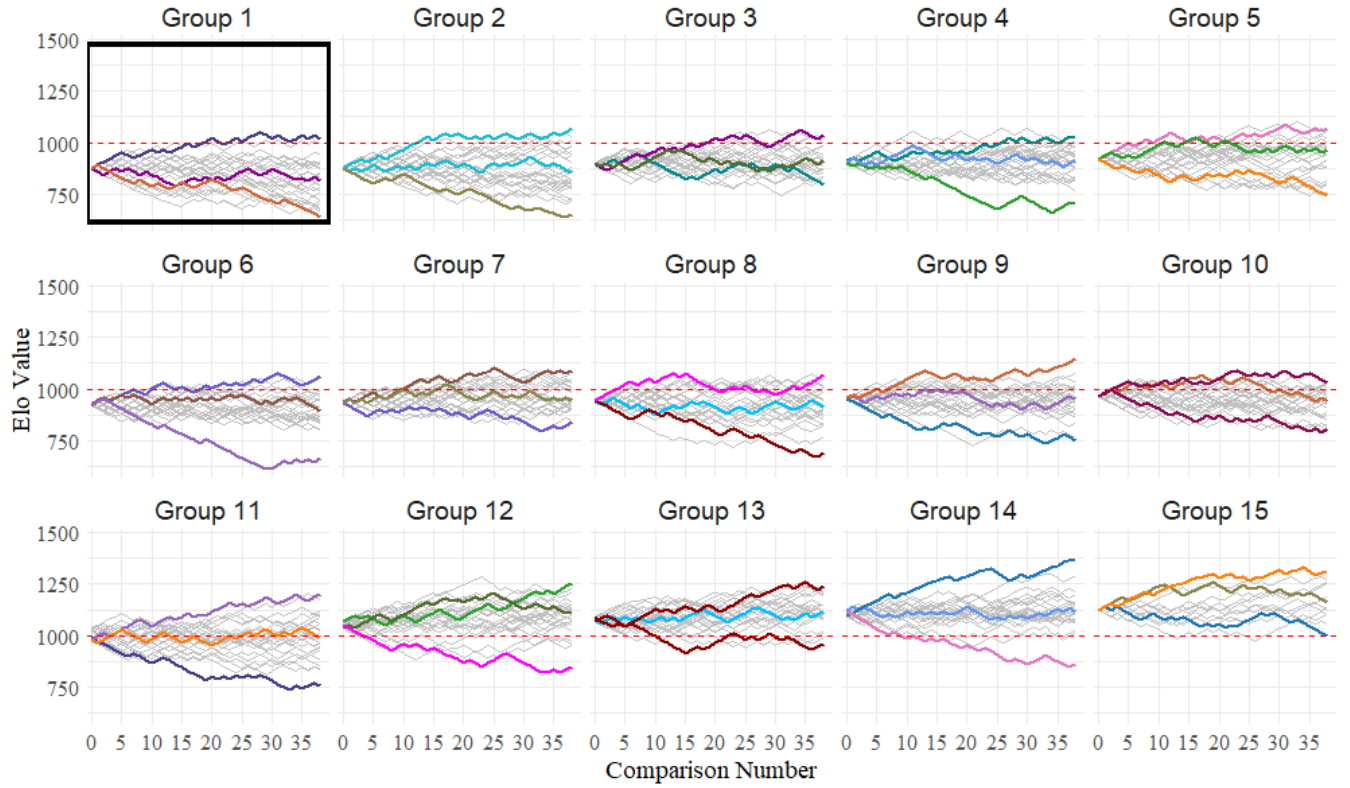


Figure 5: The evolution of statement paths. Groups are created purely for ease of visualisation purposes, where all 363 statement paths are divided across 15 groups, ordered by starting Elo score from most negative Elo start values (Group 1) to most positive start values (Group 15). Within each group the minimum, maximum, and median of the 38th comparison are highlighted for illustrative purposes. The x -axis plots the number of pairwise comparisons per statement. Group 1 is outlined to highlight this group is shown in further detail in Figure 1a.

Table 1: Regression Results for Prediction Models of the Human Elo Score After 38 Comparisons. For the human model $F(1, 361)$; for all other models $F(2, 360)$.

Model Name	Variable	Estimate	Std. Error	t value	p value	95% Confidence Interval	R ²
GPT	(Intercept)	-33.6738	40.9999	-0.8213	0.4120	[-114.30, 46.96]	0.7878
	E_0	0.3136	0.0608	5.1594	0.0000	[0.19, 0.43]	
	E_{38}	0.7212	0.0348	20.7127	0.0000	[0.65, 0.79]	
GEMINI	(Intercept)	-95.5015	43.0579	-2.2180	0.0272	[-180.18, -10.82]	0.7599
	E_0	0.3369	0.0659	5.1168	0.0000	[0.21, 0.47]	
	E_{38}	0.7619	0.0415	18.3622	0.0000	[0.68, 0.84]	
CLAUDE	(Intercept)	-19.9119	39.6279	-0.5025	0.6156	[-97.84, 58.02]	0.8027
	E_0	0.2508	0.0598	4.1912	0.0000	[0.13, 0.37]	
	E_{38}	0.7698	0.0348	22.0973	0.0000	[0.70, 0.84]	
RANDOM	(Intercept)	-226.3605	59.0950	-3.8305	0.0002	[-342.58, -110.15]	0.5368
	E_0	1.2952	0.0768	16.8633	0.0000	[1.14, 1.45]	
	E_{38}	-0.0608	0.0518	-1.1749	0.2408	[-0.16, 0.04]	
HUMAN	(Intercept)	-231.8640	58.9401	-3.9339	0.0001	[-347.77, -115.95]	0.5350
	E_0	1.2400	0.0608	20.3806	0.0000	[1.12, 1.36]	

Table 2: Top 3 Best and Worst ranked statements at the end of the task for both human and GPT-3.5-turbo after 38 comparisons.

Type	Rank	Event ID	Statement
Human Best	1st	168	<i>"Positive changes in my life coming into effect, like work, money, creative projects"</i>
	2nd	406	<i>"My family have been homeless for a while and today they finally got a house out of the blue. The council messed us about for over 5 years and finally the day came and I am over the moon."</i>
	3rd	317	<i>"my boyfriend proposed to me two days ago it was the most amazing feeling ,because I love him very much and cant wait to spend the rest of my life with him"</i>
Human Worst	1st	222	<i>"death; cheating"</i>
	2nd	409	<i>"Mother cheating on my father; Me splitting with my partner of 2 years recently; Housemates getting physically abusive"</i>
	3rd	327	<i>"A person really important and close to me passed away."</i>
GPT Best	1st	233	<i>"I had a lovely date with a girl I love. She came over to my place, and I made lunch; burgers and chips, and they were absolutely delicious. We talked, cracked jokes, and then I gave her a full-body massage. The date ended with mindblowing sex, culminating in multiple orgasms for us both. We cuddled after and hung out for a bit. We've become even closer ever since, and I think I'm gonna marry her."</i>
	2nd	405	<i>"At work, I recently received positive feedback from my boss on a project I was working on. Their appreciation for my dedication and contribution boosts my confidence and makes me feel appreciated. This positive reception also generated comfortable feedback, which motivated me to continue delivering high quality work."</i>
	3rd	14	<i>"My partner and I are currently preparing for a day at the beach, and my mum has just given me a gift of money towards my wedding so that just feels like everything is working out for me today."</i>
GPT Worst	1st	371	<i>"My brother was intensely attacked with a hockey stick by one of our very rich neighbours. All my brother did was ask them to slow down as they almost hit his dog with their car. This man assaults people with hockey sticks and threatened to kill our whole family while my infant son was asleep in the house. The man was released on bail not even 24 hours after and we live in this house in fear everyday."</i>
	2nd	322	<i>"Finding out I had cancer."</i>
	3rd	143	<i>"my mum is going into hospital to find out if the bleed on her brain is spread and therefore fatal"</i>